

Computational Finance in Python

Universität Tübingen

Dr. Thomas Schön

Summer term 2026

Regular Assessment

assessment_notebook.ipynb

Scope:

- implement 3 trading signals in pre-defined notebook
- reasonable statistics
- reasonable graphs

Requirements:

- assessment_notebook: needs to be able to run stand alone on another computer
- explanations / formulas: need to be written in LaTeX
- delivery deadline:
28.06.2026 EOD
- to thomas.schoen@schoen.global

research_notebook.ipynb module.py

Scope:

- provide empirical evidence for your trading signals and their parameters: no pre-defined notebook available
→ write your own notebook and module
- reasonable statistics
- reasonable graphs

Requirements:

- research_notebook: needs to be able to run stand alone on another computer
- module: encapsulate reusable functions and use them in both notebooks
- explanations / formulas: need to be written in LaTeX
- delivery deadline:
28.06.2026 EOD
- to thomas.schoen@schoen.global

live demo / presentation

Scope:

- your delivered Jupyter notebooks (without modifications)
- don't prepare any slides, just present your notebooks
- reading your presentation will result in a lower grade

Requirements:

- 20min presentation
- 10min defense
- assessment date: **01.07.2026**
- **attendance is mandatory for every group member**
- **free presentation without reading**

100% group work / group grading

100% individual grading

ERASMUS Assessment

assessment_notebook.ipynb

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research_notebook.ipynb module.py

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100% of course grade
100% group work

Assessment Type Overview



regular students and double-degree students
(degree seeking in Tübingen):
regular assessment only



incoming ERASMUS students:
ERASMUS assessment only



all other incoming exchange students:
regular assessment or ERASMUS assessment
(please specify in your registration)

Assessment Hints

assessment_notebook.ipynb research_notebook.ipynb module.py

NumPy only / no built-in functions of other libraries allowed

- Good code: only use NumPy functionalities for your numerical computations
- Bad code: use built-in functions of other libraries like Pandas such as `rolling(window=n).mean()` that are not transparent regarding the computational steps

Reusability of your code

- Good code: write few functions that you can reuse as often as possible, reducing the lines of code; include these functions in `module.py` and import the module into both notebooks to reuse its functions there
- Bad code: copy and paste existing code functions into multiple spots in your code, increasing the lines of code artificially; no use of `module.py`

Readability of your code

- Good code: create namings for variables and functions that are self-explanatory
- Bad code: name variables and functions in a non-obvious way

Comments in your code

- Good code: write comments for providing context that is not provided by namings
- Bad code: there are no comments in your code

live demo / presentation

Explain the economic reasoning behind your code

- Why did you implement a particular signal
- Why did you choose a specific stock / data set (in-sample / out-of-sample data sets)
- Why did you choose a particular approach to calibrating parameters

Provide convincing evidence and results

- Explain the parameters and outcomes of your strategy economically and why your strategies are beneficial to a potential investor
- Present well-formatted graphs and plots
- Present convincing parameters and tables

Full explanation within 20min presentation

- Explain both notebooks within 20min of presentation
- Focus on the above mentioned hints

Don't explain your code lines

- The quality of your code will be assessed before your presentation
- Don't explain any code lines in your presentation

hints are not exhaustive
non-mentioned aspects also make up for the grade

Assessment

- All code has to be written by the groups themselves
- Group work starts on 05.05.2026
- **You are allowed to use any AI tools**

**IN CASE OF ANY
ATTEMPTS OF DECEPTION
(„TÄUSCHUNGSVERSUCHE“)
THE GROUP PERFORMANCE
WILL BE EVALUATED WITH 5,0**

Registration

- has to include all required information (otherwise you can't participate)
- possible **only via form**: <https://forms.gle/VR4126bb12fosj5B6>
- possible until **03.05.2026 EOD**
- process is GDPR compliant

YOUR REGISTRATION IS IMMEDIATELY BINDING

**IF YOU LEAVE THE ASSESSMENT AFTER A
BINDING REGISTRATION YOUR PERFORMANCE
WILL BE EVALUATED WITH 5,0**